

The empirical recurrence for OEIS A250930: recovering a conjecture that is not written down

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Abstract

OEIS A250930 records “Empirical recurrence of order 64”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on $S = 256$ vertices and satisfies some monic recurrence of order at most S ; Berlekamp–Massey applied to $2S$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 64, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 64 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A250930 is “Number of $(n+1) \times (3+1)$ 0..3 arrays with no 2×2 subblock having a diagonal absolute difference less than its antidiagonal absolute difference.”. It has offset 1 and begins

16588, 1104364, 73855420, 4951389320, 332020435016, 22261912983384, 1492624513556120, 100078269167905748

The entry states:

Empirical recurrence of order 64 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 12:51:54, revision 6) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 256$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S = 256$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 512$ exact terms of the model returns order 64 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{64} c_i a(n-i),$$

c_1	144	c_{17}	−178471146718060278	c_{33}	3507784504333770326541	c_{49}	−628741973
c_2	−8104	c_{18}	1781361685522446417	c_{34}	11212999310017794883271	c_{50}	2072255873
c_3	270492	c_{19}	−10155512355722945285	c_{35}	−19722165561839855773884	c_{51}	−67868317
c_4	−6161630	c_{20}	42155260415671247563	c_{36}	12065547890541733379240	c_{52}	343401224
c_5	99730093	c_{21}	−138406206547966314039	c_{37}	13443485175141760257193	c_{53}	−24473306
c_6	−1121196008	c_{22}	373044358666303838266	c_{38}	−45645663952438029267485	c_{54}	159386904
c_7	7587545511	c_{23}	−838697314192850653721	c_{39}	65255313713197798434684	c_{55}	−83446258
c_8	−3961110497	c_{24}	1573168416083872457654	c_{40}	−57214623312437116092507	c_{56}	35443434
c_9	−602017133753	c_{25}	−2411869132329842910573	c_{41}	21545435531503636551260	c_{57}	−12705921
c_{10}	8269378274357	c_{26}	2821579677213712749117	c_{42}	26155358606228985040805	c_{58}	39823369
c_{11}	−65943459092676	c_{27}	−1882581067433523342854	c_{43}	−64911725026000097561093	c_{59}	−1088775
c_{12}	357133465237518	c_{28}	−1353190580387236551801	c_{44}	81058794964709757800493	c_{60}	24288073
c_{13}	−1301168295812184	c_{29}	7039627238307997770673	c_{45}	−74272352095309360518860	c_{61}	−382580
c_{14}	2682731067293063	c_{30}	−13584361303380333588711	c_{46}	54193926266586053877827	c_{62}	3181968
c_{15}	−458368241353275	c_{31}	17429408905094487552319	c_{47}	−32309076415370413821928	c_{63}	−6074
c_{16}	−1070558269083224	c_{32}	−14523007920422294655383	c_{48}	15792048493571367082180	c_{64}	−11052

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 64, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $64 + 4$ terms removed returns order 64 as well, so $\deg q_{\min} = 64 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 12 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $64 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 64 that this sequence satisfies — and that family turns out to have exactly one member.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A250930.
- [2] J. L. Massey, Shift-register synthesis and BCH decoding, *IEEE Trans. Inform. Theory* **15** (1969), 122–127.
- [3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)
- [4] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.